Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Agentic AI for Business Process Automation

TGS Ref No: TGS-2024045801

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.2

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	31 July 2026	Initial release — 2-day (14-hour) lesson plan for Agentic AI for Business Process Automation, aligned to the 5 hands-on labs and TSC AER-TEM-3026-1.1.	Dr. Alfred Ang
1.1	1 August 2026	Day 2 rescheduled so the final assessment follows instruction within the training day; daily timing now driven from a single source shared with the slide deck; assessment time stated separately from instructional time.	Dr. Alfred Ang
1.2	1 August 2026	Added a screenshot of each lab's application — the four lab websites with their chat widgets open, plus the Lab 1 n8n canvas and sample flyer — to the slide deck and the Learner Guide.	Dr. Alfred Ang

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Course Information

Course Title	Agentic AI for Business Process Automation
WSQ Course Reference	TGS-2024045801
TSC Title / Code	Artificial Intelligence Application · AER-TEM-3026-1.1
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7 training hours per day (14 hours)
Daily Timing	Day 1: 9:30 am – 6:00 pm. Day 2: 9:00 am – 5:30 pm instruction, assessment to 7:45 pm. A 1-hour lunch and two 15-minute tea breaks, additional to instructional time.
Mode	Instructor-led, hands-on workflow-automation labs built in n8n
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain what n8n and AI agents are, and build automated workflows using triggers and actions.
- LO2: Implement AI agent applications that are triggered by webhooks and integrated with business systems.
- LO3: Design agentic chatbot workflows that operate safely within business and regulatory constraints.
- LO4: Apply human-in-the-loop approval so that sensitive actions are authorised by a person.
- LO5: Deploy Retrieval-Augmented Generation (RAG) to ground an agent in an organisation's own documents.

Skills Framework Alignment

This course is mapped to the TSC Artificial Intelligence Application (AER-TEM-3026-1.1).

Code	Ability / Knowledge statement
A1	Implement AI applications in collaboration with technology service providers
A2	Deploy AI workflows according to plan
A3	Identify and report any issues with the AI applications and data collected
K1	Background and basics of AI
K2	Fundamental concepts and methods of statistics and programming for data science
K3	Mathematics and computing theories
K4	Machine learning techniques and applications
K5	Principles of data management
K6	Software development methodologies
K7	Types of AI applications

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K8	AI implementation procedures

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on n8n build tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The Written Assessment covers the Knowledge (K) statements; the Practical Performance covers the Ability (A) statements.
- Final assessment is conducted at the end of Day 2.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — n8n, AI Agents and Agentic Chatbots

Time	Duration	Topic / Activity (with slide references)
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, course outline, learning outcomes, Skills Framework alignment and mandatory digital attendance (AM). Slides 2–19
10:00–11:00	60 min	**Foundation — Why agentic AI: the evolution of AI, agentic vs generative AI, from RPA to Agentic Process Automation, business value and use cases. Slides 20–28**
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	**Topic 1 — Overview of n8n and AI Agents: what n8n is, hosting options, credentials, nodes, triggers and actions, JSON and expressions, execution modes. Slides 29–37**
12:30–13:30	60 min	Lunch break
13:30–14:15	45 min	**Topic 1 (continued) — What is an AI agent: model, system instruction, memory and tools; webhooks as the front door; structured output. Slides 38–42**
14:15–15:30	75 min	Hands-on Lab 1 — Lab 1 — n8n Trigger and Action (Event Flyer with QR Code). Slides 44–60
15:30–15:45	15 min	Tea break
15:45–17:30	105 min	Hands-on Lab 2 — Lab 2 — Webhook and AI Agent (Marina

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		Trust Bank Onboarding). Slides 61–77
17:30–18:00	30 min	Day 1 recap, Q&A and PM digital attendance

Instructional time: 420 minutes (7 hours). Tea breaks and lunch are additional to instructional time.

Day 2 — Human in the Loop, RAG and Assessment

Time	Duration	Topic / Activity (with slide references)
9:00–9:15	15 min	Day 1 recap and mandatory digital attendance (AM)
9:15–10:15	60 min	**Topic 2 — Agentic Chatbot and Human in the Loop: what an agentic workflow is, rule-based vs agentic automation, safe chatbot design and memory. Slides 79–84**
10:15–10:30	15 min	Tea break
10:30–12:00	90 min	Hands-on Lab 3 — Lab 3 — Chatbot Webhook (Investment Advisor): the contact gate and the four compliance probes. Slides 90–105
12:00–13:00	60 min	Lunch break
13:00–13:45	45 min	**Topic 2 (continued) — Human in the loop: when a human belongs in the workflow, the approval pattern and building the accountability trail. Slides 85–88**
13:45–15:00	75 min	Hands-on Lab 4 — Lab 4 — Human in the Loop (Client Rapport Assistant): the approve and decline paths. Slides 106–124
15:00–15:15	15 min	Tea break
15:15–16:00	45 min	**Topic 3 — Retrieval-Augmented Generation: tokenization, embeddings, vector stores, chunking and grounding against hallucination. Slides 126–136**
16:00–17:15	75 min	Hands-on Lab 5 — Lab 5 — RAG Customer Care Chatbot (Cook & Bake Academy): ingestion and the grounded chatbot. Slides 138–158
17:15–17:30	15 min	Course recap, taking it back to work, Q&A. Slides 160–162
17:30–17:45	15 min	**Briefing for Assessment and Assessment digital attendance**
17:45–18:45	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
18:45–19:45	60 min	**Practical Performance (PP)**

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		— hands-on n8n build tasks, 1 hour, open book. TRAQOM survey and Assessment digital attendance

Instructional time: 420 minutes (7 hours). Assessment: 2 hours (WA 1 hour + PP 1 hour), plus a 15-minute briefing. Tea breaks and lunch are additional to instructional time.

Total instructional time across the course: 14 hours (2 days × 7 hours), plus 2 hours of assessment (WA 1 hour + PP 1 hour) and a 15-minute assessment briefing.

Lab Reference (aligned to topics and learning outcomes)

Topic	Outcomes	Labs	Deck slides
Topic 01: Overview of n8n and AI Agents	LO1 · LO2	Lab 1, Lab 2	Slides 29–78
Topic 02: Agentic Chatbot and Human in the Loop	LO3 · LO4	Lab 3, Lab 4	Slides 79–125
Topic 03: Retrieval-Augmented Generation (RAG)	LO5	Lab 5	Slides 126–159

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